

Do relationship changes cause changes in emotional well-being? A longitudinal investigation

 [sciencedirect.com/science/article/pii/S0191886926000334](https://www.sciencedirect.com/science/article/pii/S0191886926000334)

Zhencheng Fan, ..., Chun Pan

[Personality and Individual Differences](#)

[Volume 254](#), May 2026, 113670

Open access

Highlights

- •
Finds evidence that changes in relationship status cause changes in emotional wellbeing
- •
Participants experience higher emotional wellbeing when in an intimate relationship than single.
- •
Participants experience higher emotional wellbeing when in a good quality intimate relationship than single.

Abstract

The current study tests the hypothesis, derived from evolutionary theory, that relationship status constitutes a significant predictor of emotional well-being. We analyzed data from thirteen waves of the Pairfam study, a longitudinal project with a representative sample of 12,000 German participants, using mixed-model analysis. Our results indicate that participants' emotional well-being was significantly higher during waves in which they were in an intimate relationship compared to waves in which they were single. We also found that singlehood was associated with more negative emotions for men than for women, though the observed difference was small. Furthermore, participants experienced higher emotional well-being when in a good-quality intimate relationship than when they were single or in a poor- or moderate-quality relationship. Participants' emotional well-being was also higher when they were single than when they were in a poor- or moderate-quality intimate relationship. These differences were generally small, although the effect on loneliness was considerable. Our findings make a strong case that changes in relationship status, including changes in relationship quality, lead to changes in emotional well-being.

- [Previous article in issue](#)
- [Next article in issue](#)

Keywords

Singlehood

Emotional wellbeing

Happiness

Satisfaction with life

Relationship quality

1. Introduction

A considerable proportion of adults in contemporary societies are single, that is, they are not in an intimate relationship ([Adamczyk & Trepanowski, 2023](#); [Kislev, 2019](#)). For instance, one study found that 47% of adults in the United States under the age of 30 were not in a romantic relationship (Pew Research [Center, 2023](#)). The relatively high prevalence of singlehood raises the question of whether single people are better off in terms of their emotional well-being, specifically in terms of the balance of positive and negative emotions and the life satisfaction they experience, compared to people in an intimate relationship. The current study addresses this question. We begin unfolding our argument by discussing the nature of emotions.

1.1. Relationship status and emotional well-being

1.1.1. The nature of emotions

Evolutionary theorizing posits that emotions are mechanisms evolved to motivate human behavior in ways that increase the probability that the genes coding for them are represented in future generations—a concept termed fitness ([Johnston, 2000](#); [Nesse, 2019](#)). Different emotions achieve this end through distinct pathways, one being the following: Negative emotions typically punish fitness-decreasing behavior, while positive emotions reward fitness-increasing behavior ([Al-Shawaf et al., 2015](#)). In particular, these mechanisms monitor fitness outcomes, generating positive emotions—subjective states that are pleasant to experience—when individuals encounter fitness-increasing situations, and negative emotions—states that are unpleasant to experience—when individuals encounter fitness-decreasing situations. This feedback loop motivates adaptive behavior: Individuals are driven to engage in actions that increase their fitness to experience positive affect and to avoid actions that decrease fitness to escape negative affect. In this respect, the proximate goal of human behavior is to experience positive emotions and avoid negative ones; by pursuing this proximate goal, the agent ultimately behaves in a fitness-increasing manner ([Tooby & Cosmides, 2008](#)). This functional perspective is critical for understanding human psychology: Fitness-increasing behavior is often not consciously motivated by fitness-increasing logic but is instead emotionally driven.

Consider a resource acquisition example. Having available resources considerably increases fitness, as individuals can divert these resources toward their own survival and that of their kin, whereas a shortage of resources has the opposite effect. Accordingly, *ceteris paribus*, when individuals secure a job promotion associated with a substantial pay raise, their fitness potential increases. This increase triggers positive emotions, such as happiness and pride. These emotions reward the individual, who retains the memory of this affective state and takes action—such as working hard—to secure further promotions and re-experience these positive states. Conversely, when individuals lose their job, their fitness prospects decline. This triggers negative emotions, such as depression and shame, which motivate individuals to find new employment to alleviate this distress. Furthermore, the memory of this negative state serves as a deterrent; once a new job is secured, individuals take action to retain it to avoid experiencing such negative emotions again.

1.1.2. Emotions and singlehood

The evolutionary framework regarding emotion can be applied to understand the connection between emotional wellbeing and relationship status. Specifically, maintaining a long-term intimate partnership offers several fitness-increasing advantages. Primarily, humans are a

sexually reproducing species; to reproduce, individuals must gain access to the reproductive capacity of the opposite sex ([Buss, 2016](#)). Moreover, raising children to sexual maturity requires considerable, prolonged, and reliable investment, a burden often too great for a single parent. Accordingly, this endeavor is best served by effective cooperation between parents; fundamentally, to raise children successfully, humans generally require long-term mates ([Lancaster & Lancaster, 1987](#)). Long-term partners offer additional fitness benefits, including material support (e.g., financial resources) and non-material support (e.g., emotional backing, physical protection), which are particularly vital during crises, such as economic downturns ([Apostolou, Christoforou, & Lajunen, 2023](#)). Additionally, a committed long-term relationship reduces the risk of contracting sexually transmitted infections, which are more prevalent in the context of casual mating ([Tybur et al., 2011](#)).

However, the absence of an intimate partner—being single—also confers specific fitness benefits. To begin with, single individuals can focus their energy and resources on developing personal qualities that may later be used to attract higher-quality mates ([Apostolou, 2017](#)). For instance, individuals may focus on educational attainment and securing high-paying employment, attributes highly valued in the mating market. Additionally, by remaining single, individuals can concentrate efforts on addressing immediate fitness-compromising issues, such as health problems. Furthermore, by avoiding commitment to a single partner, individuals can engage in casual mating, enabling them to refine flirting skills and assess their mate value—information useful for securing a suitable long-term partner at a later stage ([Apostolou and Christoforou \(2022\)](#)).

Although temporary singlehood offers advantages contingent on life stage and context, it is reasonably argued that, *ceteris paribus*, the fitness benefits of a long-term intimate relationship generally outweigh those of singlehood ([Apostolou et al., 2023](#)). Consequently, being in an intimate relationship is typically more fitness-increasing than not being in one. Based on this premise, it can be hypothesized that emotional mechanisms are activated to motivate individuals to form lasting partnerships and avoid chronic singlehood. Specifically, being single is a potentially fitness-compromising state that triggers negative emotions, such as loneliness and sadness. These emotions motivate people to seek a partner to alleviate this distress. Conversely, being in an intimate relationship is potentially fitness-increasing, triggering positive emotions such as happiness and companionship, which motivate individuals to sustain the bond ([Apostolou et al., 2019](#)). Overall, the evolutionary framework leads to the prediction that, on average, mated individuals enjoy higher emotional wellbeing than single individuals.

1.1.3. Emotions and the quality of the relationship

While intimate relationships provide considerable fitness benefits, not all relationships are equally fitness-increasing; indeed, some may decrease fitness. For instance, while a primary benefit of a relationship is shared parental investment, some partners are unwilling or unable to provide resources, or may even extract resources from the family unit. Similarly, while a partner should ideally be a source of support, some individuals are emotionally distant, indifferent, or even physically and psychologically abusive ([Buss & Duntley, 2011](#)). In effect, individuals may find themselves in relationships that increase fitness considerably, moderately, or decrease it. Emotional mechanisms are expected to guide individuals toward pursuing fitness-increasing relationships and abandoning fitness-decreasing ones.

Specifically, when individuals find themselves in a fitness-decreasing relationship, negative emotions such as sadness and disappointment are triggered. These emotions motivate corrective action, including relationship termination, to cease the distress. The memory of this emotional state encourages caution in future mate choices to avoid similar outcomes. On the other hand, when individuals are in a fitness-increasing relationship, positive emotions such as

happiness and satisfaction are triggered, motivating the continuation of the bond. This emotional state, when remembered, motivates single individuals to seek high-quality relationships to re-experience these positive affects ([Apostolou et al., 2024](#)). Overall, the evolutionary theoretical framework on emotions lead to the prediction that on average people in a good quality intimate relationship would enjoy a higher emotional wellbeing than people in a poor quality intimate relationship.

1.1.4. Current literature

Research consistently demonstrates that mated individuals, particularly those who are married, report greater happiness and subjective well-being than their unmarried counterparts. For instance, in a seminal cross-cultural study involving nations ranging from collectivist to individualist societies, [Diener et al. \(2000\)](#), found that married individuals generally reported higher subjective well-being than divorced, separated, or never-married individuals, though the magnitude of this benefit varied by culture. Similarly, more recent analyses indicate that relationship status remains a strong predictor of life satisfaction, often mediated by factors such as relationship desire. [Kislev \(2022\)](#) highlighted that the negative impact of singlehood is frequently exacerbated by a strong desire for a relationship, suggesting that the discrepancy in well-being is not solely about status but also about unfulfilled mating goals. Furthermore, specific subgroups of singles, such as those identifying as “involuntarily celibate,” report significantly lower levels of well-being compared to both non-incel singles and mated individuals ([Costello et al., 2022](#)).

However, a methodological limitation in many of these investigations is the tendency to group diverse categories of “unmarried” individuals—including the divorced, widowed, and never-married—into a single classification. This approach can obscure the specific psychological outcomes of those who are currently single versus those in dating or committed relationships. More informative insights are derived from studies that directly compare singles to partnered individuals while accounting for psychological moderators. For example, [Girme et al. \(2016\)](#) examined a New Zealand student population and found that while individuals in intimate relationships generally reported higher life satisfaction, this effect was moderated by social goals; specifically, single individuals who prioritized avoidance social goals (trying to avoid conflict or rejection) suffered lower well-being, whereas those with approach goals were often just as happy as their partnered peers.

Cross-cultural evidence further supports the link between mating status and emotional outcomes. [Apostolou et al. \(2019\)](#) analyzed a Greek sample to explore the “price” of involuntary singlehood, finding that single participants experienced significantly fewer positive emotions, more negative emotions, and lower life satisfaction relative to those in a relationship or married. Building on this, a subsequent large-scale study comprising 6338 participants from 12 countries found that individuals in relationships or marriages reported higher scores across multiple dimensions of well-being—including positive affect, life satisfaction, optimism, and sense of purpose—compared to single participants ([Apostolou et al., 2024](#)). Crucially, this study also highlighted the role of relationship quality: Participants in good-quality intimate relationships enjoyed the highest emotional well-being, whereas those in poor-quality relationships experienced the lowest well-being, comparable to, and in some metrics worse than, participants who were involuntarily single.

Finally, longitudinal research provides compelling evidence regarding the directionality of these effects, moving beyond cross-sectional associations. Using data from the German Socio-Economic Panel (Pairfam), a representative household study in Germany, [Krämer et al. \(2025\)](#) modeled the effects of life events on satisfaction over time. They found that forming a new romantic partnership was associated with a distinct increase in life satisfaction, whereas

separation from an intimate partner precipitated a significant decline. Corroborating these findings, [Qin et al. \(2025\)](#) assessed life satisfaction and loneliness in a sample of 3165 singles at two time points, six months apart. Their analysis revealed that individuals who transitioned from singlehood to a romantic relationship reported a small but significant increase in life satisfaction and a moderate, significant decrease in loneliness, suggesting that relationship formation directly contributes to emotional improvements in specific domains.

1.1.5. The present study

Cross-sectional studies provide strong support for the hypothesis that relationship status predicts emotional well-being, with individuals in a relationship, on average, experiencing better emotional well-being than those who are single. If changes in relationship status do cause changes in emotional well-being, then mated and single people would differ in this dimension, which is what cross-sectional studies typically find. However, the design of these studies is limited for establishing causal relationships. It is also plausible that emotional well-being predicts relationship status. For instance, people who are depressed may be less likely to look for mates or to be appealing as mates, and thus more likely to be single. If this is the case, mated individuals would be less depressed than single individuals, which is also what cross-sectional studies find.

This limitation can be addressed with a longitudinal design that examines how people's emotional wellbeing changes as their relationship status changes over time. A longitudinal approach shifts the focus from between-person differences to within-person change, providing stronger evidence on whether changes in relationship status result in changes in emotional well-being. To the best of our knowledge, only two studies have adopted such a design. [Krämer et al. \(2025\)](#) examined life satisfaction without however, considering other emotional responses. [Qin et al. \(2025\)](#) also examined loneliness, but only across two waves, six months apart. Neither study examined the impact of the quality of the relationships. The purpose of the current study is to extend this literature by examining the effect of relationship status on emotional well-being across a broader set of emotions, taking also into consideration the quality of an intimate relationship, over waves that span several years.

2. Methods

2.1. Data

The present study was based on thirteen waves of the Pairfam (Panel Analysis of Intimate Relationships and Family Dynamics) study ([Brüderl et al., 2021](#); [Huinink et al., 2011](#)), which is a large-scale, long-term longitudinal research project in Germany. It is designed to investigate the development of partnership and family life in Germany over an extended period. The study began in 2008 with a nationally representative sample of over 12,000 young adults from three birth cohorts: 1971–1973, 1981–1983, and 1991–1993. The survey employs a fixed-panel design, in which the same ‘anchor’ participants were interviewed annually, allowing for the analysis of intra-individual change over time. A refreshment sample was drawn at the start of wave 11. The goal of the wave 11 refreshment was to increase the number of anchor respondents. Currently, there are 14 waves; however, the 14th wave did not include measurements for several variables of interest, and therefore, it was not included in our analysis. The demographic information for each wave is summarized in Table A in the Supplementary section. Full demographic information is available from the project's official website.

2.2. Measures

Life satisfaction and relationship satisfaction were measured on a 0–10 scale (0, very dissatisfied; 10, very satisfied). The positive and negative emotions assessed in the study are summarized in [Table 1](#), [Table 2](#), respectively. Participants indicated how frequently they felt each emotion using the following scale: 1, almost never; 2, sometimes; 3, often; 4, almost always.

Table 1. Fixed effects of relationship status on positive emotions and life satisfaction.

	Relationship status						Relationship status * Sex	
	Empty Cell	Empty Cell	In a relationship	Empty Cell	Married	Empty Cell	Empty Cell	Empty Cell
	F	p-Value	b	p-Value	b	p-Value	F	p-Value
Life satisfaction	922.39	<0.001	0.42 (0.02)	<0.001	0.86 (0.03)	<0.001	12.87	<0.001
Women	442.76	<0.001	0.41 (0.02)	<0.001	0.79 (0.03)	<0.001		
Men	517.31	<0.001	0.50 (0.02)	<0.001	0.84 (0.03)	<0.001		
I enjoy life	92.88	<0.001	0.05 (0.01)	<0.001	0.14 (0.01)	<0.001	6.81	0.001
Women	41.91	<0.001	0.05 (0.01)	<0.001	0.12 (0.01)	<0.001		
Men	61.56	<0.001	0.08 (0.01)	<0.001	0.14 (0.01)	<0.001		
I am happy	593.38	<0.001	0.18 (0.01)	<0.001	0.25 (0.01)	<0.001	2.70	0.067
I feel good	117.42	<0.001	0.08 (0.01)	<0.001	0.13 (0.01)	<0.001	4.33	0.013
I feel secure	158.23	<0.001	0.11 (0.01)	<0.001	0.25 (0.01)	<0.001	31.13	<0.001
Women	126.69	<0.001	0.11 (0.01)	<0.001	0.22 (0.01)	<0.001		
Men	49.85	<0.001	0.06 (0.01)	<0.001	0.14 (0.02)	<0.001		

	Relationship status					Relationship status * Sex		
	Empty Cell	Empty Cell	In a relationship	Empty Cell	Married	Empty Cell	Empty Cell	Empty Cell
	F	p-Value	b	p-Value	b	p-Value	F	p-Value
Positive emotions								
I am calm and composed	4.59	0.010	−0.02 (0.01)	0.038	−0.01 (0.01)	0.493	5.25	0.005

Note. The reference category for relationship status is 'single'.

Table 2. Fixed effects of relationship status on negative emotions.

	Relationship status					Relationship status * Sex		
	Empty Cell	Empty Cell	In a relationship	Empty Cell	Married	Empty Cell	Empty Cell	Empty Cell
	F	p-Value	b	p-Value	b	p-Value	F	p-Value
Negative emotions								
I feel lonely	2056.68	<0.001	−0.55 (0.01)	<0.001	−0.73 (0.02)	<0.001	40.89	<0.001
Women	752.75	<0.001	−0.54 (0.01)	<0.001	−0.68 (0.02)	<0.001		
Men	1420.17	<0.001	−0.69 (0.01)	<0.001	−0.94 (0.02)	<0.001		
I am sad	233.08	<0.001	−0.10 (0.01)	<0.001	−0.19 (0.01)	<0.001	17.55	<0.001
Women	106.83	<0.001	−0.09 (0.01)	<0.001	−0.16 (0.01)	<0.001		
Men	146.55	<0.001	−0.13 (0.01)	<0.001	−0.16 (0.01)	<0.001		
I am depressed	180.70	<0.001	−0.10 (0.01)	<0.001	−0.18 (0.01)	<0.001	10.35	<0.001
Women	88.75	<0.001	−0.09 (0.01)	<0.001	−0.15 (0.01)	<0.001		
Men	104.26	<0.001	−0.11 (0.01)	<0.001	−0.16 (0.01)	<0.001		

	Relationship status						Relationship status * Sex	
	Empty Cell	Empty Cell	In a relationship	Empty Cell	Married	Empty Cell	Empty Cell	Empty Cell
	F	p-Value	b	p-Value	b	p-Value	F	p-Value
Negative emotions								
I am in desperation	87.29	<0.001	−0.07 (0.01)	<0.001	−0.14 (0.01)	<0.001	19.58	<0.001
Women	48.37	<0.001	−0.06 (0.01)	<0.001	−0.11 (0.01)	<0.001		
Men	50.36	<0.001	−0.07 (0.01)	<0.001	−0.09 (0.01)	<0.001		
My mood is melancholy	157.21	<0.001	−0.07 (0.01)	<0.001	−0.17 (0.01)	<0.001	9.96	<0.001
Women	81.09	<0.001	−0.07 (0.01)	<0.001	−0.15 (0.01)	<0.001		
Men	88.35	<0.001	−0.10 (0.01)	<0.001	−0.15 (0.01)	<0.001		
My mood is gloomy	129.03	<0.001	−0.06 (0.01)	<0.001	−0.16 (0.01)	<0.001	11.17	<0.001
Women	57.05	<0.001	−0.05 (0.01)	<0.001	−0.13 (0.01)	<0.001		
Men	88.55	<0.001	−0.10 (0.01)	<0.001	−0.15 (0.01)	<0.001		

Note. The reference category for relationship status is 'single'.

Based on existing categories, we created a three-level relationship-status variable: Single, in a relationship, and married. Participants were classified as single if they did not have an intimate partner. This category included participants who were never married, divorced, or widowed. Participants were classified as in a relationship if they had an intimate partner but were not married. This category could include participants who were never married, divorced, or widowed. Participants were classified as married if they were married, regardless of cohabitation status.

Furthermore, following [Apostolou et al. \(2024\)](#), we classified intimate relationships as good or poor quality, based on subjective relationship satisfaction. In particular, we created another relationship status variable with four levels (poor, moderate, good, single). Participants who reported scores that ranged from "0" to "3" were classified as being in a poor-quality relationship, from "4" to "6" as being in a moderate-quality relationship, and from "7" to "10" as being in a good-quality relationship. Participants who were not in an intimate relationship were classified as single.

2.3. Data analysis

We used multilevel fixed-effects models to compare changes in emotional well-being over time across different relationship statuses. These models examine within-person change by comparing each participant's well-being to their own mean, which controls for selection effects due to stable individual characteristics. Each emotion and life satisfaction served as dependent variables in separate models, with relationship status and sex as independent variables, and age as a covariate. The analysis estimated the fixed effect of relationship status, which indicates how the dependent variable changes when participants transition from one relationship status to another. We conducted the analysis in two stages. Initially, we examined the effect of relationship status (single, in a relationship, or married) on positive emotions, negative emotions, and life satisfaction. Subsequently, we examined the effect of relationship status reflecting relationship quality (poor, moderate, good, or single) on positive and negative emotions and life satisfaction.

Specifically, regarding positive emotions, we performed six tests, one for each emotion. When a significant interaction between relationship status and sex was observed, we conducted the analysis for that specific emotion separately for male and female participants. This occurred for three emotions; consequently, six additional tests were performed, resulting in a total of 12 tests (Table 1). We repeated this procedure using the relationship status variable reflecting relationship quality, resulting in a total of eight tests (Table 3). Regarding negative emotions, six tests were performed for the pooled sample and 12 additional tests were conducted separately for each sex, yielding a total of 18 tests (Table 2). Regarding the impact of relationship quality on negative emotions, 12 tests were performed (Table 4). In summary, 20 tests were performed for positive emotions and 30 for negative emotions, resulting in a total of 50 tests.

Table 3. Fixed effects of relationship quality on positive emotions and life satisfaction.

	Relationship status								Relationship status * Sex	
	Empty Cell	Empty Cell	Poor	Empty Cell	Moderate	Empty Cell	Good	Empty Cell	Empty Cell	Empty Cell
Positive emotions	F	p-Value	b	p-Value	b	p-Value	b	p-Value	F	p-Value
Life satisfaction	1204.84	<0.001	-0.14	<0.001	-0.02	0.493	0.65	<0.001	0.78	0.503
I enjoy life	422.51	<0.001	-0.10	<0.001	-0.14	<0.001	0.12	<0.001	0.07	0.975
I am happy	1050.31	<0.001	-0.05	0.003	-0.09	<0.001	0.26	<0.001	3.32	0.019
I feel good	445.43	<0.001	-0.11	<0.001	-0.12	<0.001	0.15	<0.001	4.07	0.007
I feel secure	371.92	<0.001	-0.07	<0.001	-0.10	<0.001	0.20	<0.001	26.52	<0.001
Women	301.04	<0.001	-0.08	<0.001	-0.11	<0.001	0.18	<0.001		
Men	113.58	<0.001	-0.05	0.016	-0.12	<0.001	0.11	<0.001		

	Relationship status								Relationship status * Sex	
	Empty Cell	Empty Cell	Poor	Empty Cell	Moderate	Empty Cell	Good	Empty Cell	Empty Cell	Empty Cell
		p-Value	b	p-Value	b	p-Value	b	p-Value	F	p-Value
Positive emotions	F									
I am calm and composed	149.82	<0.001	-0.17	<0.001	-0.17	<0.001	0.02	0.046	1.26	0.287

Note. The reference category for relationship status is 'single'.

Table 4. Fixed effects of relationship quality on negative emotions.

	Relationship status								Relationship status * Sex	
	Empty Cell	Empty Cell	Poor	Empty Cell	Moderate	Empty Cell	Good	Empty Cell	Empty Cell	Empty Cell
		p-Value	b	p-Value	b	p-Value	b	p-Value	F	p-Value
Negative emotions	F									
I feel lonely	1925.21	<0.001	-0.21	<0.001	-0.11	<0.001	-0.70	<0.001	11.71	<0.001
Women	903.29	<0.001	-0.18	<0.001	-0.08	<0.001	-0.68	<0.001		
Men	1114.05	<0.001	-0.36	<0.001	-0.33	<0.001	-0.81	<0.001		
I am sad	453.93	<0.001	0.08	<0.001	0.07	<0.001	-0.17	<0.001	7.48	<0.001
Women	302.55	<0.001	0.09	<0.001	0.08	<0.001	-0.15	<0.001		
Men	192.52	<0.001	-0.01	0.642	0.33	0.019	-0.16	<0.001		
I am depressed	374.57	<0.001	0.05	0.001	0.06	<0.001	-0.16	<0.001	2.69	0.045
I am in desperation	335.49	<0.001	0.13	<0.001	0.07	<0.001	-0.13	<0.001	13.66	<0.001
Women	247.58	<0.001	0.15	<0.001	0.09	<0.001	-0.11	<0.001		
Men	122.23	<0.001	0.06	<0.001	0.07	<0.001	-0.10	<0.001		
My mood is melancholy	311.34	<0.001	0.36	0.015	0.05	<0.001	-0.13	<0.001	1.16	0.324

	Relationship status						Relationship status * Sex			
	Empty Cell	Empty Cell	Poor	Empty Cell	Moderate	Empty Cell	Good	Empty Cell	Empty Cell	Empty Cell
		p-Value	b	p-Value	b	p-Value	b	p-Value	F	p-Value
Negative emotions	F									
My mood is gloomy	319.93	<0.001	0.10	<0.001	0.10	<0.001	-0.13	<0.001	2.82	0.038

Note. The reference category for relationship status is ‘single’.

3. Results

3.1. Descriptives

Means and standard deviations were calculated for all levels of relationship status for the pooled sample (see Table B, Supplementary material). For positive emotions and life satisfaction, participants gave higher scores when in a relationship or married than when single. In addition, they gave higher scores when in a good intimate relationship, than when single, than when in poor or moderate quality relationship. With respect to negative emotions, participants reported lower scores when married or in a relationship than when single. Moreover, they reported lower negative emotions when in good quality intimate relationship, than when they were single, than when they were in a poor or moderate quality intimate relationship.

3.2. Relationship status

In total, we conducted 50 statistical tests. To reduce Type I error inflation, we applied a Bonferroni correction, setting the familywise alpha to 0.001 across 50 tests (0.05/50). The fixed effects of relationship status on positive emotions and life satisfaction are summarized in [Table 1](#). All effects were significant except for “I am calm and composed.” The coefficients indicate that when participants were in a relationship or married, they experienced more positive emotions and greater life satisfaction than when they were single. For life satisfaction, the coefficient for the “in a relationship” category was 0.42, meaning that, on average, an individual's life satisfaction was predicted to be 0.42 points higher during waves when they were in a relationship than during waves when they were single. The difference was more than twice as large for married participants compared to single participants. Given the 0–10 scale used to measure life satisfaction, these differences can be interpreted as small or small-to-moderate. For “I am happy,” the coefficient for “married” was 0.25, indicating that an individual's happiness was predicted to be 0.25 points higher during waves when they were married than during waves when they were single. Given the 1–4 scale, this difference can be considered as moderate. For life satisfaction and “I feel secure,” there was a significant interaction between relationship status and sex, indicating that the effect of relationship status on these outcomes differed by sex. For life satisfaction, the difference between being single and being in an intimate relationship was slightly larger for men than for women, although the coefficients were very close. For the “I enjoy life,” the interaction was also significant, with men scoring slightly higher than women. This pattern was further observed for the “I am happy” and “I feel good,” although the interaction was not significant. For “I feel secure,” the difference was more pronounced, with

coefficients approximately twice as large for women as for men, indicating that being in an intimate relationship versus being single was associated with a larger increase in perceived security among women than among men.

The fixed effects of relationship status on negative emotions are summarized in [Table 2](#). In all cases, the effect was significant, with negative coefficients indicating that participants experienced fewer negative emotions when they were in a relationship or married than when they were single. For “I feel lonely,” the coefficient for the “married” category was -0.73 , meaning that, on average, an individual's loneliness was predicted to be 0.73 points lower during waves when they were married than during waves when they were single. Given the scale used, this constitutes a large difference. By contrast, the differences were small for the other negative emotions. Additionally, in all cases, there was a significant interaction between relationship status and sex. The observed differences tended to be larger for men than for women.

3.3. Relationship quality

With respect to relationship status reflecting quality of the relationship, as we can see from [Table 3](#), it was significant for all positive emotions and life satisfaction. In waves participants were in a poor or moderate-quality relationship, they experienced fewer positive emotions and lower life satisfaction than when they were single. On the other hand, when participants were in a good quality intimate relationship, they experienced more positive emotions than when single. In all instances, the differences were relatively small. For example, for life satisfaction, an individual's life satisfaction was predicted to be 0.14 units lower in waves they were in a poor-quality intimate relationship than in waves they were single. In addition, a participant's life satisfaction was not predicted to be different when in a moderate quality intimate relationship than when single. On the other hand, participants were predicted to be 0.65 units higher when in a good quality intimate relationship than when single. A significant interaction between relationship status and sex was found only of the “I feel secure”, where the effect was more pronounced for women than for men in poor quality and moderate quality relationships versus single.

As we can see from [Table 4](#), relationship status was significant for all negative emotions. With respect to loneliness, we can see that in waves participants were in an intimate relationship, irrespective of its quality, they were less lonely than when single. For instance, participants were predicted to be 0.70 units less lonely when they were in a good intimate relationship than when single, which given the four-points scale used, can be interpreted to be a considerable difference. In addition, there was a significant interaction between relationship status and sex, with the observed effect being more pronounced for male than for female participants. On the other hand, for the rest of the negative emotions, when participants were in a poor or moderate quality intimate relationship, they experienced more negative emotions than when single, with the pattern reversing when they were in a good quality intimate relationship. In all instances, the differences were relatively small. There was a significant interaction between relationship status and sex for the “I am sad”, where male participants as opposed to female participants were not predicted to differ when in a poor and moderate quality intimate relationship than when single. There was also a significant such interaction for the “I am in desperation,” where the effect was less pronounced for men than for women.

4. Discussion

In the current study, we investigated the impact on people's emotional well-being when their relationship status changed. We found that transitioning from being single to being in a relationship was associated with higher emotional well-being, including fewer negative emotions

(such as loneliness) and more positive emotions (such as happiness), as well as increased life satisfaction. In addition, we found that the quality of an intimate relationship mattered. When people were in a good intimate relationship, they experienced a higher emotional wellbeing than when they were single, with the pattern reversing when they were in a poor or moderate quality intimate relationship. This evidence provides strong support for the hypothesis that relationship status is a significant determinant of emotional well-being.

Our findings suggest that there is an impact of relationship status on emotional well-being. This impact is generally small, consistent with previous studies ([Apostolou et al., 2024](#); [Qin et al., 2025](#)). Emotional mechanisms track fitness, which is influenced by numerous factors other than relationship status, such as health, income, career, and close friendships ([Dunbar, 2023](#)). For these mechanisms to motivate behavior toward fitness-increasing ends, each factor should have a relatively small effect. For instance, if having a strong social network made people extremely happy, the happiness system would not be able to reward them for achieving other fitness-increasing goals, such as obtaining a job promotion, because happiness could not increase further. In this respect, it is not surprising that relationship status had only a small effect on emotional well-being.

By contrast, for specific emotions, such as loneliness, the effect was considerable, aligning with previous findings ([Apostolou et al., 2024](#); [Qin et al., 2025](#)). One reason may be that the loneliness mechanism evolved specifically to motivate individuals to seek the company of others in general, and of intimate partners in particular. Unlike broader emotional mechanisms, such as happiness, it monitors a narrow, fitness-relevant dimension: Whether someone is alone or in the company of others. Accordingly, the absence of social contact is likely to result in high levels of loneliness. This is consistent with our finding that participants experienced less loneliness when in an intimate relationship, irrespectively of its quality, than when single.

For several emotions, we observed a significant interaction between relationship status and sex, indicating that relationship status affects the emotional wellbeing and life satisfaction of men and women differently. From an evolutionary perspective, this suggests that the fitness implications of being single versus being in an intimate relationship differ between the sexes. The general trend indicated that the disparity between singlehood and partnership was more pronounced for men; specifically, being single was associated with higher levels of negative emotions, lower levels of positive emotions, and lower life satisfaction for men than for women. This suggests that singlehood may exact a higher psychological and potential fitness cost on men. One explanation is that men typically engage in more intensive intrasexual competition for mate access, effectively placing women in the position of selectors ([Buss, 2016](#)). Consequently, *ceteris paribus*, women may find it comparatively easier to secure mates; thus, prolonged singlehood may pose a greater risk to the reproductive fitness of men than to that of women. An exception to this general trend was observed regarding feelings of security: Single women reported feeling less secure than single men. A plausible evolutionary explanation is that, historically, women often relied on intimate partners for physical protection, leading to a selected predisposition to seek mates who can provide such security ([Buss, 2016](#)). Consequently, the absence of an intimate partner may be perceived as more consequential for the safety—and historically, the fitness—of women than of men along this specific dimension.

Our findings strongly suggest that the quality of an intimate relationship, rather than relationship status itself, is a key determinant of a person's emotional well-being. Participants who transitioned into a good-quality intimate relationship experienced the highest levels of well-being, whereas those who entered a poor or moderate-quality relationship experienced the lowest. This finding is consistent with cross-sectional, cross-cultural research, which shows that individuals in good-quality relationships report the greatest emotional well-being, while those in poor-quality

relationships report the least ([Apostolou et al., 2024](#)). We would also like to note that, across waves, only about 16% of participants indicated that they were in a poor- or moderate-quality intimate relationship. This pattern does not reflect a sample bias but rather the tendency for individuals to end relationships they perceive as low quality. Consequently, poor- and moderate-quality relationships are less frequent than good-quality ones.

The longitudinal design of the current study strengthens the case for a causal relationship between relationship status and emotional well-being. However, there are considerations that prevent a definitive causal claim. In particular, our analysis did not control for time-varying confounders, that is, variables that change over time along with relationship status and life satisfaction. For instance, individuals may receive a major promotion and salary increase. This greater financial security and success could make people happier and more satisfied with their lives, as well as more desirable as mates, and thus less likely to be single.

Additional limitations need also to be considered. To begin with, singles are not a homogeneous group, with some singles choosing to be single, others single because they face difficulties in attracting an intimate partner, and others who are currently between relationships. These subgroups differ in their emotional well-being ([Girme et al., 2022](#); [Kislev, 2022](#)). The current study did not differentiate among types of singles, which limits our ability to estimate the effect of relationship status on emotional well-being. Future studies should address this limitation by using a more refined measure of relationship status. We also need to consider that the dataset was confined to the German cultural context, so our findings may not generalize to other contexts. Accordingly, future studies should attempt to replicate our findings in different cultural settings.

In summary, evolutionary theorizing indicates that changes in relationship status bring changes in emotional well-being. Using a longitudinal design, the current study offers strong evidence in support of this hypothesis. Future studies should attempt to provide further evidence by replicating this finding in different cultural settings and by using more refined measures of relationship status.

CRediT authorship contribution statement

Menelaos Apostolou: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Elyakim Kislev:** Writing – review & editing, Methodology, Investigation, Conceptualization.

Declaration of competing interest

None.

Appendix A. Supplementary data

Data availability

The authors do not have permission to share data.

References

1. [Adamczyk and Trepanowski, 2023](#)

K. Adamczyk, R. Trepanowski

Singlehood in Europe: Rates and factors associated with happiness

V&R unipress (2023)

2. [Al-Shawaf, Conroy-Beam, Asao and Buss, 2015](#)

L. Al-Shawaf, D. Conroy-Beam, K. Asao, D.M. Buss

Human emotions: An evolutionary psychological perspective

3. [Apostolou, 2017](#)

M. Apostolou

Why people stay single: An evolutionary perspective

4. [Apostolou, Alexopoulos and Christoforou, 2023](#)

M. Apostolou, S. Alexopoulos, C. Christoforou

The price of being single: An explorative study of the disadvantages of singlehood

5. [Apostolou and Christoforou, 2022](#)

M. Apostolou, C. Christoforou

What makes single life attractive: An explorative examination of the advantages of singlehood

6. [Apostolou, Christoforou and Lajunen, 2023](#)

M. Apostolou, C. Christoforou, T.J. Lajunen

What are romantic relationships good for? An explorative analysis of the perceived benefits of being in a relationship

7. [Apostolou, Matogian, Koskeridou, Shialos and Georgiadou, 2019](#)

M. Apostolou, I. Matogian, G. Koskeridou, M. Shialos, P. Georgiadou

The price of singlehood: Assessing the impact of involuntary singlehood on emotions and life satisfaction

8. [Apostolou et al., 2024](#)

M. Apostolou, M. Sullman, A. Błachnio, O. Burýšek, E. Bushina, F. Calvo, ..., S. Font-Mayolas

Emotional wellbeing and life satisfaction of singles and mated people across 12 nations

9. [Bröderl et al., 2021](#)

J. Bröderl, S. Drobnič, K. Hank, B. Johannes Huinink, F.J.N. Nauck, S. Walper, ..., B. Wilhelm

The German Family Panel (pairfam). GESIS Data Archive, Cologne. ZA5678 Data file Version 12.0. 0

10. [Buss, 2016](#)

D.M. Buss

The evolution of desire: Strategies of human mating

(4th ed.), Basic Books (2016)

11. [Buss and Duntley, 2011](#)
D.M. Buss, J.D. Duntley

The evolution of intimate partner violence
12. [Costello, Rolon, Thomas and Schmitt, 2022](#)
W. Costello, V. Rolon, A.G. Thomas, D. Schmitt

Levels of well-being among men who are incel (involuntarily celibate)
13. [Diener, Gohm, Suh and Oishi, 2000](#)
E. Diener, C.L. Gohm, E. Suh, S. Oishi

Similarity of the relations between marital status and subjective well-being across cultures
14. [Dunbar, 2023](#)
R. Dunbar

Friends: Understanding the power of our most important relationships

(2023)
15. [Girme, Overall, Faingataa and Sibley, 2016](#)
Y.U. Girme, N.C. Overall, S. Faingataa, C.G. Sibley

Happily single: The link between relationship status and well-being depends on avoidance and approach social goals
16. [Girme, Park and MacDonald, 2022](#)
Y.U. Girme, Y. Park, G. MacDonald

Coping or thriving? Reviewing intrapersonal, interpersonal, and societal factors associated with well-being in singlehood from a within-group perspective
17. [Huinink et al., 2011](#)
J. Huinink, J. Brüderl, B. Nauck, S. Walper, L. Castiglioni, M. Feldhaus

Panel analysis of intimate relationships and family dynamics (pairfam): Conceptual framework and design

Zeitschrift für Familienforschung - Journal of Family Research, 23 (1) (2011), pp. 77-101
18. [Johnston, 2000](#)
V.S. Johnston

Why we feel: The science of human emotions

Basic Books (2000)
19. [Kislev, 2019](#)
E. Kislev

Happy singlehood: The rising acceptance and celebration of solo living

University of California Press (2019)

20. [Kislev, 2022](#)

E. Kislev

Relationship desire and life satisfaction among never-married and divorced men and women

21. [Krämer, Rohrer, Lucas and Richter, 2025](#)

M.D. Krämer, J.M. Rohrer, R.E. Lucas, D. Richter

Life events and life satisfaction: Estimating effects of multiple life events in combined models

22. [Lancaster and Lancaster, 1987](#)

J.B. Lancaster, C.S. Lancaster

The watershed: Change in parental investment and family-formation strategies in the course of human evolution

J.B. Lancaster, J. Altmann, A.S. Rossi, L.R. Sherrod (Eds.), Parenting across the life span: Biosocial dimensions, Aldine Publishing (1987), pp. 187-205

23. [Nesse, 2019](#)

R.M. Nesse

Good reasons for bad feelings: Insights from the frontier of evolutionary psychiatry

Dutton (2019)

24. [Pew Research Center, 2023](#)

Pew Research Center

For valentine's day: 5 facts about single Americans

25. [Tooby and Cosmides, 2008](#)

J. Tooby, L. Cosmides

The evolutionary psychology of the emotions and their relationship to internal regulatory variables

M. Lewis, J.M. Haviland-Johnes, L.F. Barrett (Eds.), Handbook of emotions (3rd ed.), Guilford (2008), pp. 114-137

26. [Tybur, Bryan, Magnan and Hooper, 2011](#)

J.M. Tybur, A.D. Bryan, R.E. Magnan, A.E. Hooper

Smells like safe sex: Olfactory pathogen primes increase intentions to use condoms

Cited by (2)

- [Is Parenthood Contributing to Emotional Wellbeing? The Neutrality Paradox and a Possible Resolution](#)
2026, Evolutionary Psychology
- [The study of singlehood: from a marginalized relationship status to an](#)

[emerging social category](#)

2026, Journal of Gender Studies

© 2026 The Authors. Published by Elsevier Ltd.